

C-TRACE User Role Training Dataset for AI Assistant

SECTION 1: PROJECT MANAGEMENT BASICS

Q1: Add a Verifier to a Project

User Query:

How do I add a verifier to my project?

Assistant Answer:

To add a verifier, go to 'My Projects' and select your project. Open the 'Verifiers' tab and click 'Add Verifier'. Fill in the verifier's name, organization, email, and accreditation ID. Select the verifier type (e.g., baseline, periodic, or soil lab), then click 'Send Invite'. The verifier will appear in the list with 'Pending' status.

Q2: Upload Baseline Evidence

User Query:

How do I upload baseline data for my project?

Assistant Answer:

Open your project from 'My Projects'. Go to the 'Baseline Data' section and click 'Add Baseline Evidence'. Upload geotagged photos, drone or satellite images, the project GIS map, and your baseline survey report. Enter survey date, team name, and any notes, then click 'Submit Baseline'. Your files will be shown in the list with timestamps.

Q3: Submit Monitoring Update

User Query:

How do I submit a monitoring update?

Assistant Answer:

Go to 'My Projects', open your project, then click 'Monitoring Updates' or 'Progress'.

Select 'Add New Update', choose the reporting period, upload your field photos and drone images, enter number of saplings and survival rate, add notes, and click 'Submit Update'. Your update will be listed with a submission date.

Q4: Upload KYC or Project Document

User Query:

How do I upload a document for my project?

Assistant Answer:

Open your project, go to the 'Documents' tab, and click 'Upload Document'. Select document type (KYC, Land NOC, Consent, Lab Report, Audit, etc.), choose your file, add a title and description, and click 'Upload'. The document will appear with type, upload date, and uploader name.

Q5: View Project Status and Pending Tasks

User Query:

How can I check my project status and pending tasks?

Assistant Answer:

On the dashboard, look at the 'My Projects' table. The 'Status' column shows each project's stage (Draft, Baseline Submitted, Under Verification, Verified, Credits Issued). Click a project name to open details and the 'Tasks' or 'Checklist' section to see and complete any pending actions.

Q6: See Issued and Available Credits

User Query:

Where do I see my issued carbon credits?

Assistant Answer:

Open your project and go to the 'Credits' or 'Carbon Credits' tab. Here you'll find total issued, retired, and available credits. Click on a batch to see the issuance date, verifier, and verification report.

Q7: Register a New Blue Carbon Project

User Query:

How do I register a new project?

Assistant Answer:

On the dashboard, click 'Projects' and select 'Create New Project'. Fill out project name, description, ecosystem type, location (with coordinates), land details, area, number of saplings, and methodology. Upload your land permissions and declarations, then click 'Submit Project'. You'll receive your project confirmation and Project ID.

Q8: Complete User/Organization Profile

User Query:

How do I set up my user profile?

Assistant Answer:

After logging in, go to 'My Profile'. Enter organization name, registration type and ID, address, and contact details. Upload required KYC documents and click 'Save Profile'. Your profile status will show as 'Under Review' or 'Verified' upon approval.

Q9: Remove/Replace a Verifier

User Query:

How do I remove or change a verifier on my project?

Assistant Answer:

Open your project, go to the 'Verifiers' tab. Find the verifier, click the options icon or 'Remove'. Confirm in the prompt. To replace, click 'Add Verifier' and follow the steps for new assignment.

Q10: General Navigation and Help

User Query:

What are the main sections for a user in C-TRACE?

Assistant Answer:

The main user sections are: Dashboard, My Projects, Baseline Data, Monitoring Updates, Verifiers, Documents, Credits, and Profile. Each section is accessible from the left sidebar or top navigation bar.

SECTION 2: FIELD DATA & IMAGE MANAGEMENT

Q11: VIEW PROJECT MANGROVE IMAGES WITH LOCATION MAP

User Query:

How can I see photos of my mangrove project uploaded by field workers along with their GPS locations on a map?

Assistant Answer:

Go to 'My Projects' and open your project. Click on the 'Gallery' or 'Project Images' tab. You'll see all photos uploaded by field workers displayed with their exact GPS coordinates on an interactive map. Each photo thumbnail shows: upload date, GPS location (latitude/longitude), field worker name, and photo notes. Click any marker on the map to view the photo and details. You can filter images by date range, zone, or status. Photos with GPS tags appear on the CesiumJS map showing mangrove plantation areas. Zoom in/out to see coverage by location. This helps verify field work progress and mangrove survival across different zones.

Q12: Download GPS-Tagged Photos from Field Workers

User Query:

Can I download field worker photos with GPS information?

Assistant Answer:

Yes. Go to 'Monitoring Updates' and select the update containing field photos. Click 'Download All' or select individual photos. The system preserves GPS metadata in the downloaded images. When you view downloaded photos on your device, open 'Photo Details' or 'Properties' to see latitude, longitude, altitude, and timestamp. This GPS data proves the photos were taken at the correct project location and helps with verification.

Q13: Filter Mangrove Images by Zone or Location

User Query:

How do I view photos from a specific zone or area of my project?

Assistant Answer:

In the 'Gallery' tab, look for filter options at the top. Click 'Filter by Zone' and select your desired zone (e.g., 'Zone A - North Sector'). The map and gallery will update to show only photos from that zone. You can also filter by date range, field worker name, or survival status. This helps you compare mangrove growth across different project areas and identify which zones need more maintenance.

Q14: Upload Soil Lab Report (SOC Test)

User Query:

How do I upload a soil lab report for my project?

Assistant Answer:

Go to your project and open the 'Documents' tab. Click 'Upload Document', select 'Soil Lab Report (SOC)' from the dropdown. Choose your soil analysis PDF (from an accredited lab), add a title like 'SOC Report – Zone 1, 2025', optionally add the lab name and testing date in the description field, and click 'Upload'. The report will appear in the documents list and will be reviewed by the assigned verifier.

Q15: Track Individual Mangrove Plant Growth Through Photos

User Query:

Can I see how individual mangrove saplings have grown over time using photos?

Assistant Answer:

Yes. In the 'Gallery' tab, click on a specific GPS location where mangroves were planted. The system shows a timeline of all photos taken at that location sorted by date. Scroll through the timeline to see your mangroves grow from saplings to mature trees. You can also compare photos side-by-side by selecting two dates. The verifier uses this photo timeline to calculate canopy coverage and survival rate. This visual proof is valuable for credit verification.

Q16: View Monitoring Timeline and History

User Query:

Can I see all my past monitoring submissions in one place?

Assistant Answer:

Yes. Open your project and go to 'Monitoring Updates' or 'Progress'. You'll see a timeline or table of all submitted updates organized by date. Each entry shows the month/quarter, files uploaded, key metrics (saplings planted, survival rate), and submission status. Click any entry to view details or download uploaded photos.

Q17: Create Project Proposal (PDD)

User Query:

What is a Project Design Document (PDD) and how do I upload it?

Assistant Answer:

A PDD describes your project in detail: location, ecosystem type, methodology, carbon sequestration estimate, timeline, and budget. Go to your project's 'Documents' tab, click 'Upload Document', select 'Project Proposal (PDD)', upload your detailed document (PDF), add a brief description, and click 'Upload'. The PDD is reviewed during project approval.

Q18: Add Project Stakeholders

User Query:

How do I add community members and partners to my project?

Assistant Answer:

When creating a new project, fill in the 'Stakeholders' section with names and roles of community leaders, contractors, and partner organizations. You can also edit an existing project, go to the 'Stakeholders' or 'Team' section, and click 'Add Stakeholder'. Enter their name, organization, role (e.g., 'Community Leader', 'Contractor'), and contact info, then save.

SECTION 3: DATA SUBMISSION & VERIFICATION

Q19: Modify Project Details After Submission

User Query:

Can I edit my project details after submitting it?

Assistant Answer:

Yes, if your project is in 'Draft' status, go to 'My Projects', click the project, and select 'Edit'. Modify name, description, location, area, or timeline as needed, then click 'Save'. Once the project is 'Baseline Submitted' or under 'Verification', contact admin support or your assigned verifier to request changes, as some fields become locked.

Q20: Understand Carbon Credit Status

User Query:

What do the credit statuses like 'Minted', 'Retired', and 'Reserved' mean?

Assistant Answer:

'Minted' credits are newly issued after verifier audit and are available for sale or retirement. 'Retired' credits have been used for carbon offset by a buyer and are permanently closed. 'Reserved' credits are held in a buffer (5-10%) for projects that lose survival rate due to natural disasters. You can see all statuses in the 'Credits' tab.

Q21: Download Verification Report

User Query:

How do I download the verification report for my project?

Assistant Answer:

Open your project and go to 'Credits' or 'Verification Reports'. Find the report linked to your credits and click 'Download' or the document icon. The PDF will show the verifier's audit findings, field evidence review, survival rate calculation, carbon absorption estimate, and final approval status. This is your proof for credit legitimacy.

Q22: Upload Drone or Satellite Images

User Query:

What format should drone and satellite images be in?

Assistant Answer:

Drone and satellite images should be in standard formats: JPEG, PNG, or GeoTIFF. They must include GPS coordinates and timestamp metadata. When uploading in 'Baseline Data' or 'Monitoring Updates', select the images and add notes like 'Drone overflight – [Date], altitude 50m, coverage Zone A'. The system will extract GPS data and display the images on the project map.

Q23: Track Project Area on Map

User Query:

Can I see my project boundaries and area on a map?

Assistant Answer:

Yes. Open your project and look for the 'Map' or 'GIS View' section. It displays your project's GPS coordinates and boundaries using CesiumJS or similar mapping tool. You can see the project footprint, uploaded drone images marked by location, and monitoring progress visualized spatially. This helps verify coverage and identify any gaps.

Q24: Invite Another Team Member to Your Project

User Query:

How do I add another staff member from my NGO to manage the project?

Assistant Answer:

Go to your project and open 'Team' or 'Collaborators'. Click 'Add Team Member', enter their email, name, and role (e.g., 'Project Manager', 'Data Collector', 'Administrator'). Select their permissions level (view-only, edit, or full access). Click 'Send Invite'. They will receive an email and can join after accepting. You can manage permissions for each team member.

Q25: Resolve Verifier Queries or Comments

User Query:

What do I do if a verifier sends me a query or comment about my project?

Assistant Answer:

Open your project and check the 'Verifier Comments' or 'Messages' section. You'll see any questions or requests from the verifier. Read the comment and reply with clarification or additional documents. Upload any requested evidence in the 'Documents' tab. The verifier will be notified and can then proceed with their audit or request follow-up information.

Q26: Submit Plantation Photos with GPS Tags

User Query:

How do I ensure my field photos have GPS and timestamp information?

Assistant Answer:

Use your smartphone or a GPS-enabled camera. Before uploading photos to C-TRACE, confirm they include GPS metadata (latitude, longitude, altitude). You can check this in the photo's 'Details' or 'Properties' on your device. Upload photos directly in 'Monitoring Updates' or 'Baseline Data'. The system will read and display GPS data on the project map.

Q27: Check Project Approval Status

User Query:

How long does project approval take and where can I see the status?

Assistant Answer:

Go to 'My Projects' and find your project. The 'Status' column shows current approval stage. 'Draft' means awaiting your submission. 'Baseline Submitted' means waiting for admin review. 'Under Verification' means a verifier is conducting audit (typically 1-4 weeks). 'Verified' means approved and ready for credit issuance. 'Credits Issued' means active and credits are available. You can click the project for timeline details.

Q28: Prepare Required Data Before Project Registration

User Query:

What documents and information do I need before registering a project?

Assistant Answer:

Prepare: (1) Organization KYC documents (PAN, registration, GST). (2) Project location details (GPS coordinates, state, district, village). (3) Land ownership proof or NOC. (4) Project scope (area in hectares, saplings count, methodology, timeline). (5) Baseline data (photos, drone images, soil quality info). (6) Stakeholder list. (7) Signed declaration that project is not registered elsewhere. Having these ready will speed up registration.

SECTION 4: MONITORING & TRACKING

Q29: Understand Baseline vs Monitoring Data Difference

User Query:

What is the difference between baseline data and monitoring updates?

Assistant Answer:

Baseline data is collected before plantation begins and shows the initial condition of the land (vegetation, soil quality, existing mangroves). Monitoring updates are submitted regularly (monthly or quarterly) after planting to show growth, survival rate, and maintenance work. Both are required: baseline establishes the starting point; monitoring tracks progress over time.

Q30: Submit Financial Declaration

User Query:

Do I need to share financial information about my project?

Assistant Answer:

Yes, optional financial data may be requested. Go to 'Documents' and upload a 'Financial Declaration' form with estimated project cost, funding sources, and planned credit sale/retirement budget. This helps demonstrate project viability and transparency. Click 'Upload' and ensure the document is clear and professional. This information is encrypted and only shared with authorized admin.

Q31: Access Real-Time Dashboard Analytics

User Query:

What information does the user dashboard show?

Assistant Answer:

Your dashboard displays: total number of active projects, total saplings planted, total area covered, total carbon credits issued, projects pending verification, and urgent pending tasks. You'll also see a quick-access menu to create new projects, upload monitoring data, or view verifier feedback. Widgets show project status breakdown and credit balance.

Q32: Export Project Data or Reports

User Query:

Can I download my project data as a report?

Assistant Answer:

Yes. Open your project, look for 'Export' or 'Download Report' button (usually in the top-right corner). Select the format (PDF, Excel) and data to include (baseline, monitoring, credits, verification report). Click 'Export'. The system will generate and download your comprehensive project report, suitable for stakeholder presentations or compliance audits.

Q33: Link Multiple Plots to One Project

User Query:

Can I manage multiple plantation plots under one project?

Assistant Answer:

Yes. During project creation or editing, you can add multiple plots with different locations within the same ecosystem area. Each plot has its own coordinates, area, and monitoring timeline. In 'Monitoring Updates', you can upload data separately for each plot or zone. The system aggregates data for overall project carbon calculation.

Q34: Request Verifier Re-Check or Appeal

User Query:

What if I disagree with a verifier's findings?

Assistant Answer:

If you disagree, go to 'Verifier Comments' and select 'Request Review' or 'Appeal'. Provide a detailed explanation with supporting evidence (photos, measurements, lab reports) that contradicts the verifier's findings. Submit your response. The verifier will review your appeal and either accept your evidence or clarify their decision. If unresolved, you can escalate to admin.

Q35: Update User Account and Permissions

User Query:

How do I change my password or email address?

Assistant Answer:

Go to 'My Profile' at the top-right menu. Click 'Account Settings' or 'Security'. Here you can update email, phone, and password. Enter your current password, then the new password (if changing), and click 'Save Changes'. For major changes like organization details, go to 'Organization Profile' and update, then click 'Submit for Re-Verification'.

Q36: Generate Blockchain Hash for Data Integrity

User Query:

How do I verify that my uploaded data is stored immutably on blockchain?

Assistant Answer:

After submitting any baseline, monitoring, or verification data, go to 'Project Details' and look for 'Blockchain Hash' or 'Immutable Record'. Each submission gets a unique blockchain transaction ID. Click to view the hash proof. This confirms your data is tamper-proof and permanently recorded on the blockchain, ensuring transparency and preventing fraud.

Q37: Mobile App Upload for Field Workers

User Query:

Can field workers in my community upload data from their phones?

Assistant Answer:

Yes. C-TRACE has a mobile app (Android/iOS) for Workers, Panchayats, and NGOs. Field staff can download the app, log in with NGO credentials, and upload geotagged photos directly from the field with GPS and timestamp. Photos auto-sync when internet is available or saved offline and synced later. Data is consolidated in the web portal under your project.

Q38: Understanding Carbon Credit Pricing

User Query:

How are carbon credits valued and what price can I expect?

Assistant Answer:

Carbon credit value depends on methodology (Verra, Gold Standard, National), ecosystem (mangroves typically highest), and market demand. C-TRACE shows the current credit price in the 'Credits' or 'Marketplace' section. Historical rates range from INR 300-1000+ per credit. Your verifier will calculate sequestration using approved formulas. Credits minted after audit are listed at current market rate for potential buyers.

Q39: Compliance with National Standards

User Query:

Does my project need to follow national carbon standards?

Assistant Answer:

Yes. C-TRACE aligns projects with India's Carbon Credit Trading Scheme (CCTS) and National Carbon Registry (NCCR). During registration, select your preferred methodology (e.g., national CCTS, Verra VCS, Gold Standard). Your verifier will audit against that standard. All approved projects are registered with NCCR to prevent double-counting and ensure compliance with international standards (IPCC guidelines).

Q40: Troubleshooting Upload Errors

User Query:

What should I do if my file upload fails?

Assistant Answer:

If upload fails: (1) Check file size (max 50 MB per file). (2) Ensure file format is supported (PDF, JPEG, PNG, GeoTIFF). (3) Check your internet connection. (4) Clear browser cache and try again. (5) Use a different browser if error persists. If still failing, go to 'Support' or 'Help', open a ticket with the error message, and admin will assist.

SECTION 5: ADVANCED FEATURES

Q41: Understanding Survival Rate Calculation

User Query:

How is survival rate calculated and reported?

Assistant Answer:

Survival rate is the percentage of saplings that survive after planting. You estimate it based on field observations (count live vs dead trees in a sample area). Submit survival rate percentage in 'Monitoring Updates' (e.g., 85%). Your verifier validates via satellite imagery (canopy density analysis) and site visit. Survival rate affects carbon credit calculation: higher survival = more credits issued.

Q42: Reporting Environmental Issues (Floods, Pests)

User Query:

How do I report if my plantation area is affected by floods or pest damage?

Assistant Answer:

In 'Monitoring Updates', add notes and photos documenting the damage. Upload photos showing affected areas with GPS tags. Describe the issue: "Flood on [Date] in Zone A, estimated 20% saplings lost", or "Pest damage in Zone B, recovery plan in place". Submit the update. AI flags anomalies automatically. Your verifier reviews impact and may adjust survival expectations or recommend replanting.

Q43: Access Historical Project Data

User Query:

Can I view my project data from previous years?

Assistant Answer:

Yes. Open your project and go to 'Monitoring History' or 'Timeline'. You'll see all past submissions (baseline, monthly/quarterly updates, verification reports) organized by date. Click any entry to view uploaded files, metrics recorded, and verifier comments from that period. You can also export historical data for annual reports or analysis.

Q44: Multi-Agency Digital Signature Workflow

User Query:

How are government approvals (MoEFCC, CZMA) handled?

Assistant Answer:

Multi-agency approvals (MoEFCC, State CZMA, Forest Department) are managed digitally via C-TRACE. When your project reaches 'Baseline Submitted', it is routed to relevant agencies. Each agency reviews and digitally signs (e-signature) their approval. Digital signatures are recorded as blockchain hashes for transparency. You can track approval status in 'Approvals' tab: each agency shows 'Pending', 'Approved', or 'Flagged'.

Q45: Connecting to Carbon Credit Buyers

User Query:

How can I sell my carbon credits to international buyers?

Assistant Answer:

Once credits are 'Minted' (issued), they appear in the C-TRACE Marketplace (if enabled). Buyers can view your verified project, credit quantity, and terms. They can submit purchase offers. When you accept an offer, credit ownership transfers to the buyer on blockchain. A smart contract records the transaction, updates the registry, and processes payment. Credits can also be retired (permanently offset emissions) by you.

Q46: Sector-Specific Baseline Recommendations

User Query:

What specific baseline data is needed for mangrove vs seagrass projects?

Assistant Answer:

For mangroves: baseline includes tidal zone maps, existing mangrove density (stems/hectare), soil salinity, and elevation. For seagrass: baseline includes water depth, sediment type, and seagrass species present. For salt marsh: include salinity levels and vegetation cover. Upload ecosystem-specific baseline reports along with photos and satellite images. Your verifier checks that baseline matches your ecosystem type.

Q47: Scheduling Verifier Site Visits

User Query:

How do I schedule a site visit from my assigned verifier?

Assistant Answer:

Go to 'Verifiers' tab and click on your verifier's name. You'll see a 'Schedule Visit' button or calendar widget. Select preferred dates and times for site visit. Add notes like "Best time: 9 AM - 12 PM weekdays, directions: [GPS link]". Click 'Propose Date'. Verifier receives notification, confirms availability, and calendar blocks are locked. Visit date and time appear in 'Tasks' reminder.

Q48: Accessing Satellite Monitoring Data

User Query:

Can I see satellite-based canopy monitoring for my project?

Assistant Answer:

Yes. Go to your project's 'Satellite Monitoring' or 'Remote Sensing' tab. It displays real-time satellite imagery (Sentinel-2, Landsat) showing canopy density, vegetation index (NDVI), and growth trends. The system uses AI to flag anomalies (canopy loss, floods, unusual changes). Historical satellite data is available as time-series charts. This reduces dependency on manual surveys and provides independent verification.

Q49: Community Participation and Co-Benefits Tracking

User Query:

How do I document social co-benefits and community involvement in my project?

Assistant Answer:

In 'Monitoring Updates', include a 'Co-Benefits' section. Document: number of workers employed, training provided, women participation rate, income generated for community. Upload photos of community activities and collect testimonials. Attach 'Community Reports' in the 'Documents' tab. C-TRACE tracks these social metrics alongside carbon metrics, promoting equity and measuring measurable social impact.

Q50: Final Checklist Before Submitting Project

User Query:

What is the complete checklist before I submit my project for approval?

Assistant Answer:

Before submission, confirm: (1) User profile verified with correct KYC. (2) Project details complete (name, location, coordinates, area, saplings). (3) Ecosystem type selected correctly. (4) Land NOC or permission uploaded. (5) Baseline data (photos, drone images, maps, soil report) ready. (6) Verifier(s) assigned. (7) Community consent form uploaded. (8) Project proposal (PDD) uploaded. (9) Digital declaration signed and attached. (10) All required fields marked as complete. Then click 'Submit Project'.

SECTION 6: EXTENDED FEATURES & IMAGE GALLERY

Q51: Create Custom Views of Project Photos by Location

User Query:

Can I organize photos by specific mangrove zones or plantation clusters?

Assistant Answer:

Yes. In the 'Gallery' tab, click 'Create Custom View'. Select the zones you want to group (e.g., 'North Zone', 'South Zone', 'Central Area'). The system creates a custom filtered view showing only photos from those zones. You can save this view and switch between

multiple custom views. This helps you track growth patterns by area and compare survival rates across different mangrove plantation clusters.

Q52: Share Project Photo Gallery with Stakeholders

User Query:

Can I share my project photo gallery with verifiers or buyers?

Assistant Answer:

Yes. Go to 'Gallery' and click 'Share'. Select 'Share with Verifier' or 'Generate Public Link'. If sharing with verifier, enter their email and set permissions (view-only). If generating a public link, it allows external stakeholders (buyers, government agencies) to view your mangrove photos and GPS locations. You can set expiration dates on links. The verifier/buyer sees the same interactive map with all geotagged photos.

Q53: Compare Photos Over Time for Mangrove Growth

User Query:

How can I create a time-lapse or comparison of mangrove growth from different dates?

Assistant Answer:

In the 'Gallery' tab, select 'Time-Lapse View'. Choose two dates (baseline date and current date). The system displays photos from both dates side-by-side, showing mangrove growth over time. You can also click 'Animated Timeline' to see a smooth progression of photos. This visual growth proof is powerful for verifiers and buyers to confirm successful mangrove restoration and carbon sequestration.

Q54: View Drone Footage and 360° Panorama Images

User Query:

Can I view drone videos or panoramic photos of my mangrove project?

Assistant Answer:

Yes. If field workers upload drone videos or 360° panorama photos, they appear in the 'Video Gallery' or 'Panoramic View' section. Click on any video to play it with GPS overlay showing location. For panoramic photos, you can rotate and zoom to see the full

360° view. Drone footage helps verifiers assess canopy coverage, growth patterns, and overall project health without manual site visits.

Q55: Create Photo Reports for Stakeholder Presentations

User Query:

How do I generate a photo report for stakeholder meetings or buyer presentations?

Assistant Answer:

Go to 'Gallery' and click 'Generate Report'. Select photos by date range or zone. The system creates a professional PDF report with selected photos, GPS maps, survival metrics, and growth charts. You can add your own notes and branding. Download the PDF to share in presentations with verifiers, government agencies, or carbon credit buyers. This builds trust through visual evidence.

Q56: Tag and Annotate Photos with Field Notes

User Query:

Can I add notes or labels to individual field photos in the gallery?

Assistant Answer:

Yes. Click any photo in the gallery and select 'Add Tag' or 'Edit Metadata'. Add notes like "Healthy saplings - Zone A" or "Storm damage - requires replanting". You can also tag photos with status labels (e.g., 'Excellent', 'Good', 'Needs Attention'). Tags help organize photos and provide context. Verifiers can see these tags when reviewing your evidence, speeding up the audit process.

Q57: Set Photo Verification Milestones

User Query:

Can I mark important photos as verification evidence?

Assistant Answer:

Yes. In the gallery, right-click any photo and select 'Mark as Verification Evidence' or click the star icon. These marked photos are flagged for the verifier's audit. When submitting for verification, the system highlights marked photos as key evidence of

mangrove survival and growth. This directs the verifier's attention to your strongest proof photos.

Q58: Search Photos by Metadata (Date, GPS, Worker Name)

User Query:

How do I quickly find photos uploaded on a specific date or by a specific field worker?

Assistant Answer:

In the 'Gallery' tab, use the advanced search bar at the top. You can search by: date range, GPS coordinates, field worker name, zone name, or survival status. Example: "Search for all photos by 'Ram Kumar' in 'Zone B' from October 2025". Results appear instantly with matching photos displayed on the map. This saves time finding specific evidence.

Q59: Batch Download All Project Photos with GPS Data

User Query:

Can I download all my project photos at once with location information?

Assistant Answer:

Yes. Go to 'Gallery' and click 'Download All' at the top. Select format options: organize by date, zone, or worker; include CSV metadata file with GPS coordinates. The system creates a ZIP file containing all photos plus a metadata CSV showing date, GPS, worker name, and filename. This allows offline analysis and backup of all field evidence.

Q60: Upload Community Member Photos to Gallery

User Query:

Can community members or field workers contribute photos directly to the gallery?

Assistant Answer:

Yes. In the mobile app, field workers can upload photos directly. These appear in 'Pending Photos' first. As project manager, review and approve them (or reject if low quality/wrong location). Approved photos are added to the main gallery with worker

name and upload date. This enables real-time photo collection from multiple field staff across mangrove restoration sites.

SECTION 7: ADVANCED ANALYTICS & REPORTING

Q61: Generate Canopy Coverage Map from Photos

User Query:

Can the system create a canopy coverage map from my uploaded photos and satellite data?

Assistant Answer:

Yes. Go to 'Analysis' → 'Canopy Coverage'. The system uses AI to analyze all uploaded photos and satellite imagery to estimate canopy coverage percentage. It generates a heat map showing dense mangrove coverage (green) vs sparse areas (yellow/red). This automated analysis reduces manual measurement work and provides objective survival rate estimates for carbon credit calculations.

Q62: Export Mangrove Project Data in Multiple Formats

User Query:

What file formats can I export my project data in?

Assistant Answer:

You can export project data as: (1) PDF report with photos, maps, and statistics. (2) Excel spreadsheet with monitoring data, survival rates, and photo metadata. (3) GeoJSON for importing into GIS software. (4) CSV for data analysis. (5) KML for viewing in Google Earth. Go to 'Export' and select your preferred format. The system generates and downloads the file within minutes.

Q63: Track Carbon Credits from Issuance to Retirement

User Query:

Can I see the complete history of my carbon credits from creation to sale/retirement?

Assistant Answer:

Yes. Go to 'Credits' → 'Credit Ledger'. You'll see: credit batch issuance date, verifier, methodology, number of credits, blockchain transaction ID, sale/retirement details, and final status. Click any credit batch to view: linked verification report, buyer information (if sold), or retirement confirmation (if offset). This complete audit trail proves your credits' legitimacy.

Q64: Set Up Automated Photo Upload Reminders

User Query:

Can the system remind me to upload monitoring photos on schedule?

Assistant Answer:

Yes. Go to 'Settings' → 'Monitoring Schedule'. Set your preferred monitoring frequency (monthly, quarterly, etc.) and select zones. The system sends email/SMS reminders to project managers and field workers on the scheduled date. Reminders include: due date, required data type (photos, metrics), and direct upload link. This ensures consistent monitoring and prevents missed submission deadlines.

Q65: Create Multi-Project Dashboard

User Query:

Can I view metrics across all my projects in one dashboard?

Assistant Answer:

Yes. Go to the main 'Dashboard'. You'll see aggregated metrics: total saplings across all projects, total area, combined credits issued, and overall survival rate. Each project appears as a card showing key stats. Click any project card to drill down. Use filters to compare projects or view trends. This high-level overview helps you manage multiple mangrove restoration projects efficiently.

SECTION 8: SUPPORT & TRAINING

Q66: Access Detailed Help and Video Tutorials

User Query:

Where can I find help if I'm unsure about a specific feature?

Assistant Answer:

Click the 'Help' icon (?) in the top-right corner. You'll see: FAQ section, step-by-step video tutorials, and live chat support. Videos cover: project registration, photo upload, verification process, and credit management. FAQs are searchable by keyword. If you still need help, start a live chat with support staff (available during business hours). Tutorials are in multiple languages.

Q67: Contact Admin Support for Technical Issues

User Query:

What should I do if I encounter a technical problem or bug?

Assistant Answer:

Go to 'Support' → 'Report Issue'. Describe the problem, screenshot (if possible), and your browser/device type. Submit the ticket. You'll receive a ticket number and expected response time. Support team will email you updates. For urgent issues, click 'Priority Support' and select phone call or live chat. Most issues are resolved within 24 hours.

Q68: Join Webinars and Training Sessions

User Query:

Are there training sessions for learning to use C-TRACE?

Assistant Answer:

Yes. Go to 'Learning Center' → 'Webinars'. You'll see upcoming training sessions on: project registration, photo upload best practices, verification process, and carbon credit trading. Sessions are held weekly in multiple languages. Register to attend live or watch recorded sessions. Certificates of completion are provided for training credits.

Q69: Submit Feature Requests or Feedback

User Query:

Can I suggest new features for C-TRACE?

Assistant Answer:

Yes. Go to 'Settings' → 'Feedback'. Describe your feature idea and why it would be useful. Select the category (e.g., photos, verifiers, analytics). Click 'Submit'. Your feedback goes to the development team. Popular requests are prioritized. You'll be notified if your feature is implemented in future updates.

Q70: Access C-TRACE User Community Forum

User Query:

Is there a community where I can share experiences with other NGOs?

Assistant Answer:

Yes. Click 'Community Forum' in the main menu. You'll find discussion threads on: best practices, photo upload tips, project management, and success stories. Post questions and share your experiences. Other NGOs and C-TRACE team members provide answers and advice. The community is moderated for quality. You can also connect with similar projects and learn from their strategies.

SECTION 9: COMPLIANCE & CERTIFICATIONS

Q71: Download Compliance Certificates

User Query:

Can I download certificates proving my project follows international standards?

Assistant Answer:

Yes. Once your project is 'Verified', go to 'Credentials' or 'Certificates'. Download certificates showing compliance with: IPCC guidelines, Verra standards, Gold Standard, or National CCTS. These certificates are blockchain-verified and can be shared with stakeholders, government agencies, and international buyers. They enhance your project's credibility in the carbon market.

Q72: Understand Carbon Registry Synchronization

User Query:

How are my project credits synced with the National Carbon Registry (NCCR)?

Assistant Answer:

When your project is verified, credits are automatically registered with NCCR to prevent double-counting. Go to 'Registry Status' to view: NCCR registration date, registered credit quantity, and synchronization status. The blockchain ensures two-way sync: your local data updates NCCR, and NCCR updates are reflected in C-TRACE. This synchronization is audited quarterly by independent agencies.
